**VANIER COLLEGE**

**420-521-VA - Advanced Networks - Section 00003**

# VPN

*Hosting a Virtual Private Network*

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**Instructor:** Tiago Bortoletto Vaz

**Due Date:** November 10, 2026

**Repository:** https://github.com/J0YSICK/AdvancedNetworks_VPN.git

# 1. Hardware

Homelab using an old computer running an Ubuntu OS, details to be revealed later since we don't have access to it directly in class.

# 2. Hosting

For today's lab, the Wireguard hosting was done on an SSD with a Debian OS. Our final project VPN will be done on Joy's home server.

# 3. Software

- Tailscale
- RiseupVPN
- Docker
- Wireguard

# 4. Work Done Today

Followed this setup: https://www.wireguard.com/quickstart/

Installing Wireguard:

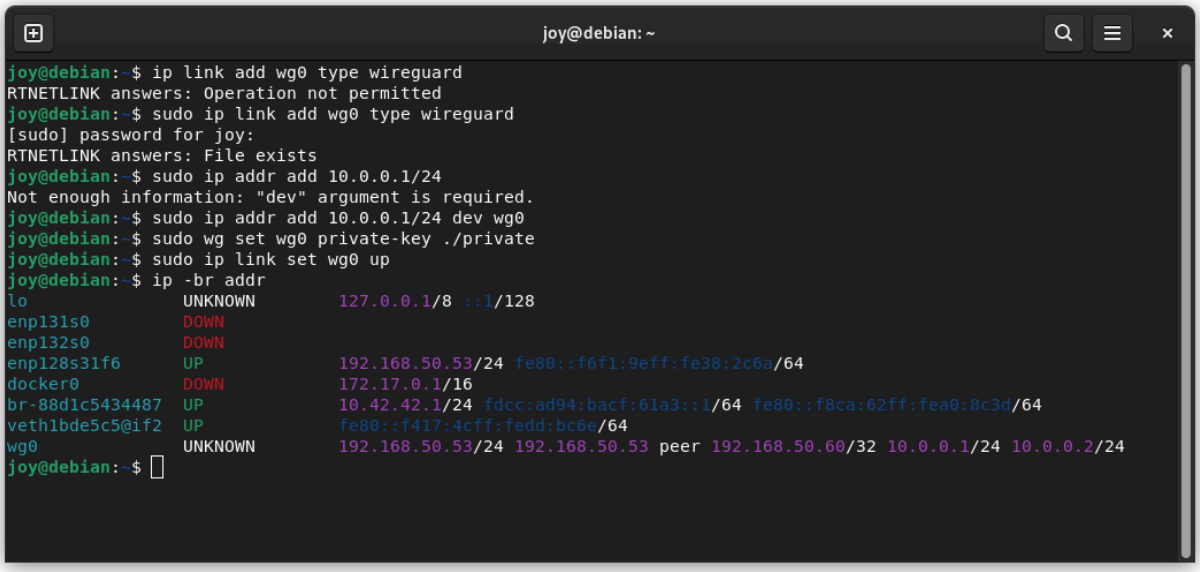
```
sudo apt install wireguard
```

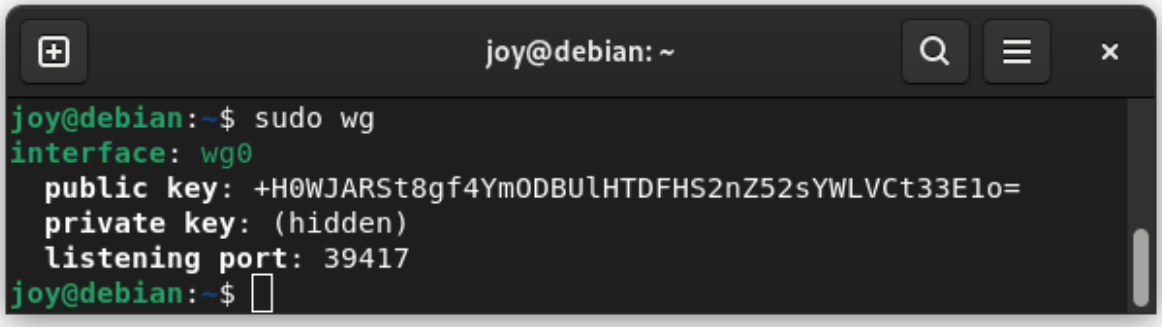
Generating public and private keys:

```
wg genkey > private
cat private
wg pubkey < private
```

Setting up interface wg0 that will act as a tunnel interface:

```
sudo ip link add dev wg0 type wireguard
sudo ip addr add 10.0.0.1/24 dev wg0
sudo wg set wg0 private-key ./private
sudo ip link set wg0 up
ip -br addr
```

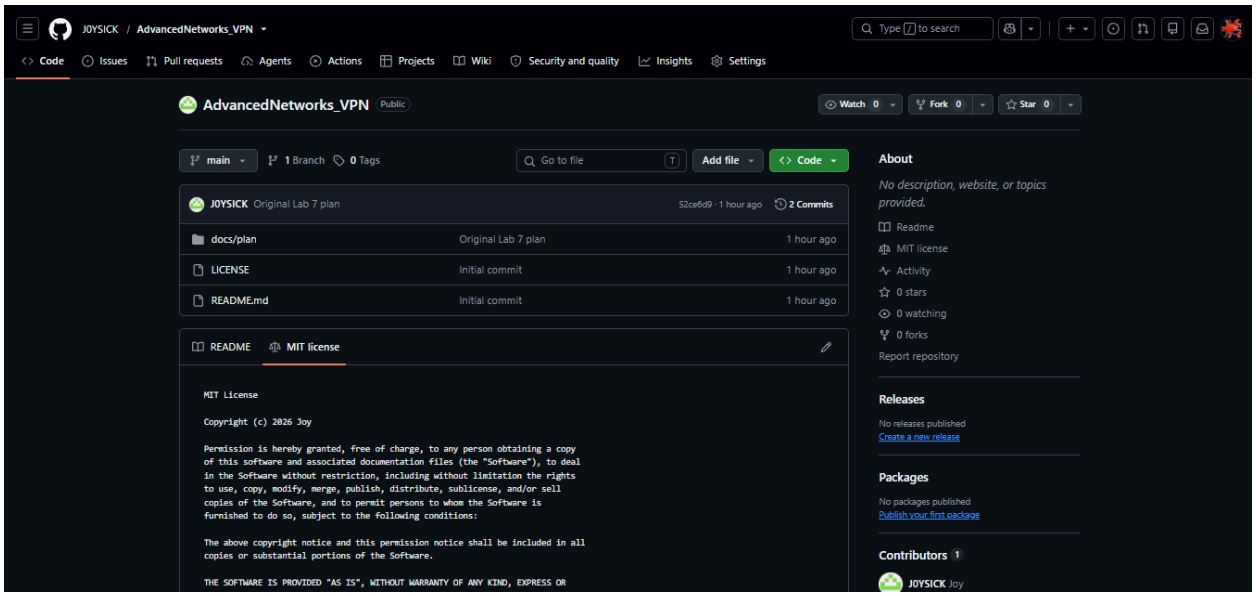



```
sudo wg set wg0 peer
+H0WJARSt8gf4YmODBUlHTDFHS2nZ52sYWLVCt33E1o= allowed-ips
10.0.0.2/32 endpoint 192.168.50.53:39417
```

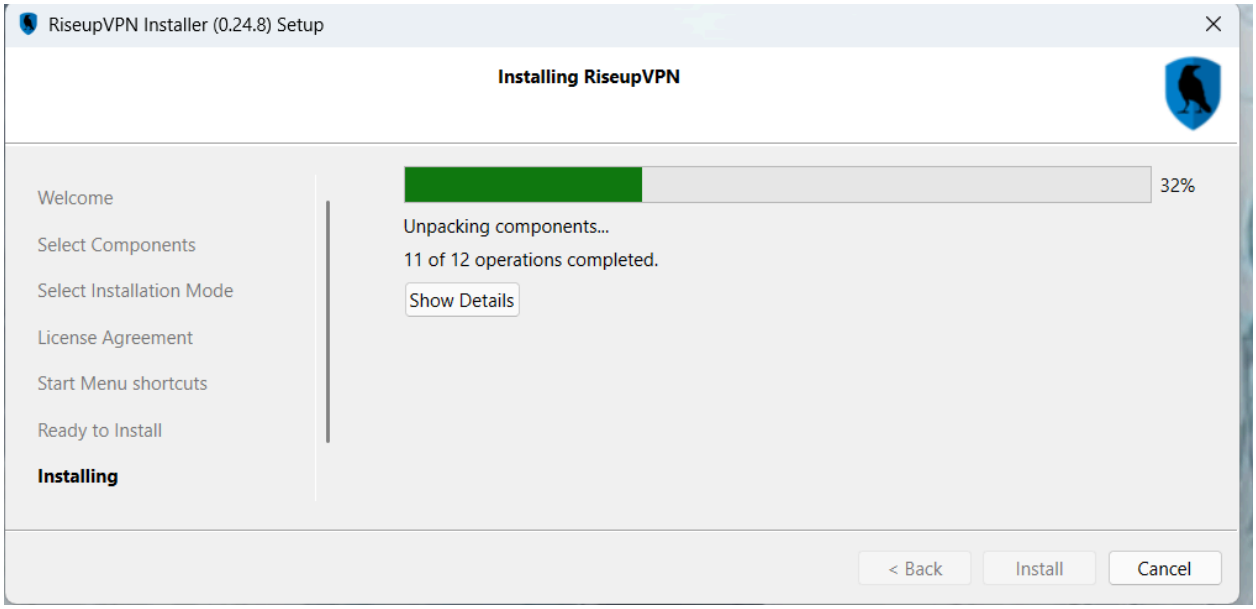
Create PowerPoint Slides

Create GitHub repository

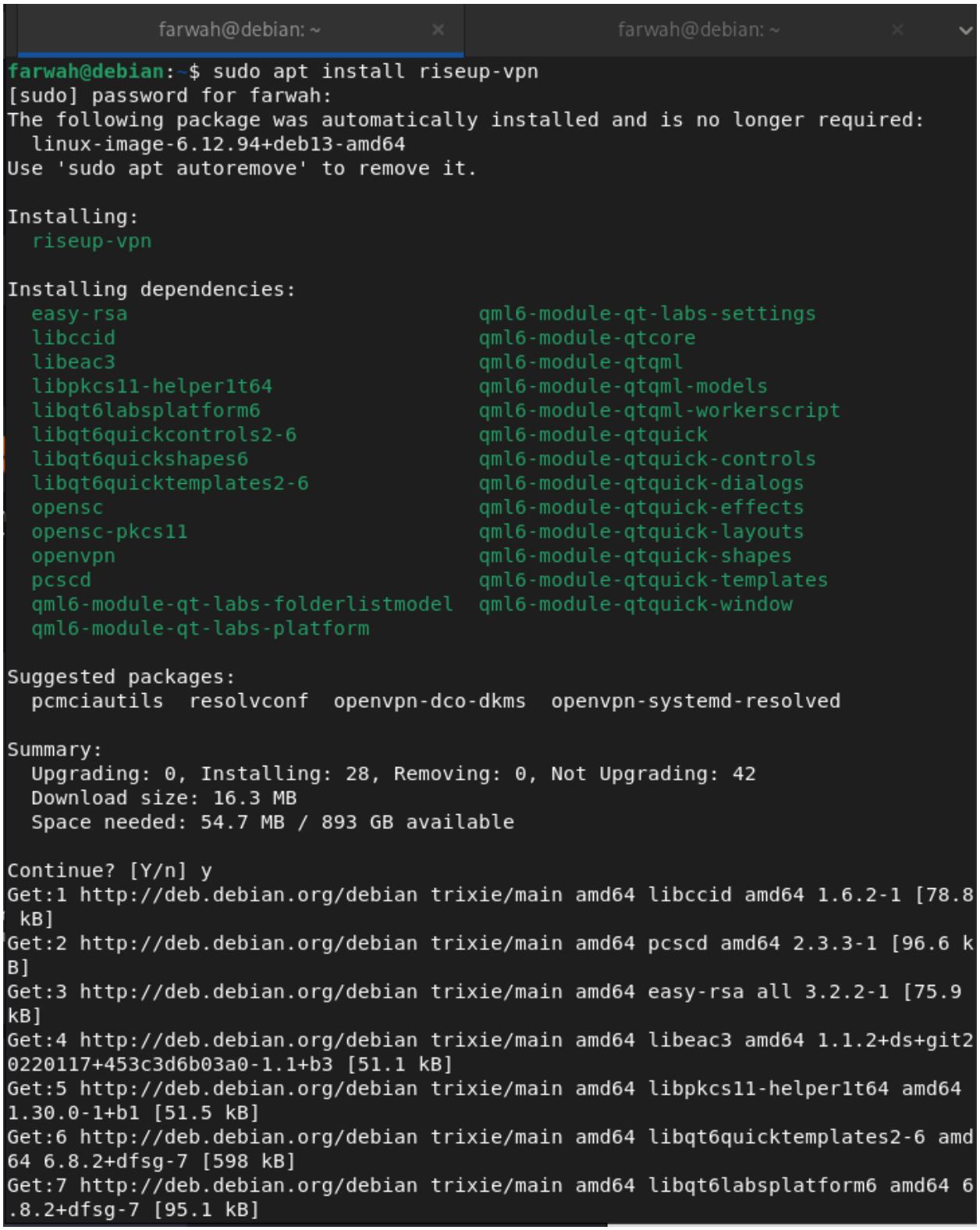


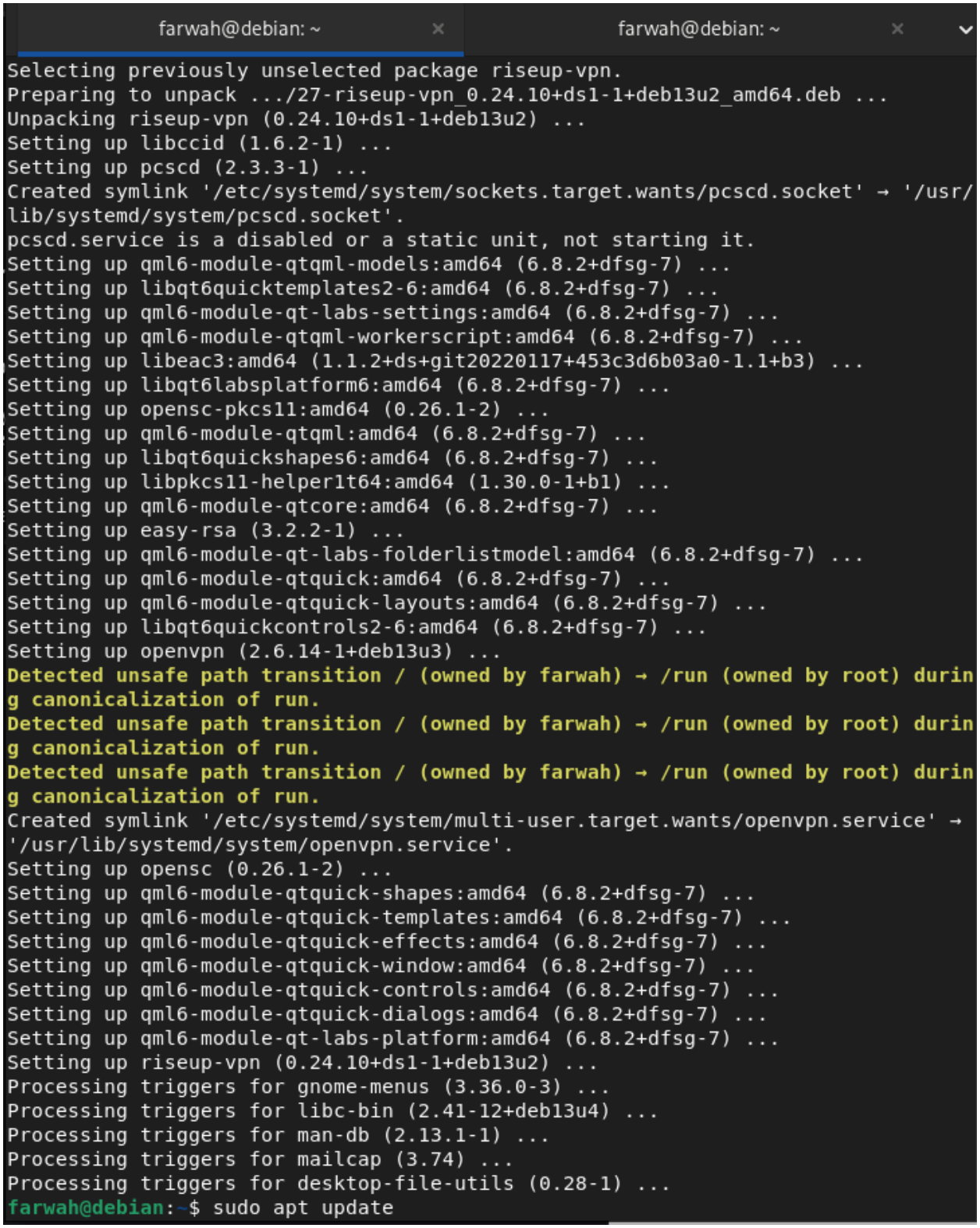

# Install and Try RiseupVPN

- Link to download in Windows: [https://riseup.net/en/windows](https://riseup.net/en/windows)



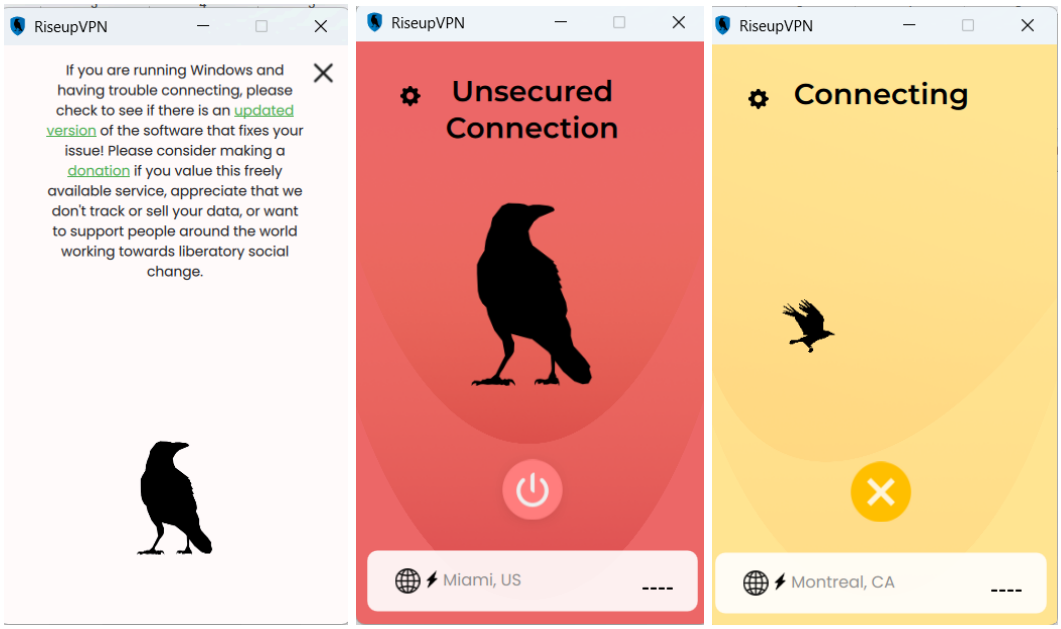

- Link to download in Linux: https://riseup.net/en/linux


farwah@debian: ~
farwah@debian: ~
farwah@debian:~$ sudo apt install riseup-vpn
[sudo] password for farwah:
The following package was automatically installed and is no longer required:
linux-image-6.12.94+deb13-amd64
Use 'sudo apt autoremove' to remove it.
Installing:
riseup-vpn
Installing dependencies:
easy-rsa
libccid
libeac3
libpkcs11-helper1t64
libqt6labsplatform6
libqt6quickcontrols2-6
libqt6quickshapes6
libqt6quicktemplates2-6
opensc
opensc-pkcs11
openvpn
pcscd
qml6-module-qt-labs-folderlistmodel
qml6-module-qt-labs-platform
qml6-module-qt-labs-settings
qml6-module-qtcore
qml6-module-qtqml
qml6-module-qtqml-models
qml6-module-qtqml-workerscript
qml6-module-qtquick
qml6-module-qtquick-controls
qml6-module-qtquick-dialogs
qml6-module-qtquick-effects
qml6-module-qtquick-layouts
qml6-module-qtquick-shapes
qml6-module-qtquick-templates
qml6-module-qtquick-window
Suggested packages:
pcmciautils resolvconf openvpn-dco-dkms openvpn-systemd-resolved
Summary:
Upgrading: 0, Installing: 28, Removing: 0, Not Upgrading: 42
Download size: 16.3 MB
Space needed: 54.7 MB / 893 GB available
Continue? [Y/n] y
Get:1 http://deb.debian.org/debian trixie/main amd64 libccid amd64 1.6.2-1 [78.8 kB]
Get:2 http://deb.debian.org/debian trixie/main amd64 pcscd amd64 2.3.3-1 [96.6 kB]
Get:3 http://deb.debian.org/debian trixie/main amd64 easy-rsa all 3.2.2-1 [75.9 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 libeac3 amd64 1.1.2+ds+git20220117+453c3d6b03a0-1.1+b3 [51.1 kB]
Get:5 http://deb.debian.org/debian trixie/main amd64 libpkcs11-helper1t64 amd64 1.30.0-1+b1 [51.5 kB]
Get:6 http://deb.debian.org/debian trixie/main amd64 libqt6quicktemplates2-6 amd64 6.8.2+dfsg-7 [598 kB]
Get:7 http://deb.debian.org/debian trixie/main amd64 libqt6labsplatform6 amd64 6.8.2+dfsg-7 [95.1 kB]


farwah@debian: ~
farwah@debian: ~
Selecting previously unselected package riseup-vpn.
Preparing to unpack .../27-riseup-vpn_0.24.10+ds1-1+deb13u2_amd64.deb ...
Unpacking riseup-vpn (0.24.10+ds1-1+deb13u2) ...
Setting up libccid (1.6.2-1) ...
Setting up pcscd (2.3.3-1) ...
Created symlink '/etc/systemd/system/sockets.target.wants/pcscd.socket' → '/usr/lib/systemd/system/pcscd.socket'.
pcscd.service is a disabled or a static unit, not starting it.
Setting up qml6-module-qtqml-models:amd64 (6.8.2+dfsg-7) ...
Setting up libqt6quicktemplates2-6:amd64 (6.8.2+dfsg-7) ...
Setting up qml6-module-qt-labs-settings:amd64 (6.8.2+dfsg-7) ...
Setting up qml6-module-qtqml-workerscript:amd64 (6.8.2+dfsg-7) ...
Setting up libeac3:amd64 (1.1.2+ds+git20220117+453c3d6b03a0-1.1+b3) ...
Setting up libqt6labsplatform6:amd64 (6.8.2+dfsg-7) ...
Setting up opensc-pkcs11:amd64 (0.26.1-2) ...
Setting up qml6-module-qtqml:amd64 (6.8.2+dfsg-7) ...
Setting up libqt6quickshapes6:amd64 (6.8.2+dfsg-7) ...
Setting up libpkcs11-helper1t64:amd64 (1.30.0-1+b1) ...
Setting up qml6-module-qtcore:amd64 (6.8.2+dfsg-7) ...
Setting up easy-rsa (3.2.2-1) ...
Setting up qml6-module-qt-labs-folderlistmodel:amd64 (6.8.2+dfsg-7) ...
Setting up qml6-module-qtquick:amd64 (6.8.2+dfsg-7) ...
Setting up qml6-module-qtquick-layouts:amd64 (6.8.2+dfsg-7) ...
Setting up libqt6quickcontrols2-6:amd64 (6.8.2+dfsg-7) ...
Setting up openvpn (2.6.14-1+deb13u3) ...
Detected unsafe path transition / (owned by farwah) → /run (owned by root) during canonicalization of run.
Detected unsafe path transition / (owned by farwah) → /run (owned by root) during canonicalization of run.
Detected unsafe path transition / (owned by farwah) → /run (owned by root) during canonicalization of run.
Created symlink '/etc/systemd/system/multi-user.target.wants/openvpn.service' → '/usr/lib/systemd/system/openvpn.service'.
Setting up opensc (0.26.1-2) ...
Setting up qml6-module-qtquick-shapes:amd64 (6.8.2+dfsg-7) ...
Setting up qml6-module-qtquick-templates:amd64 (6.8.2+dfsg-7) ...
Setting up qml6-module-qtquick-effects:amd64 (6.8.2+dfsg-7) ...
Setting up qml6-module-qtquick-window:amd64 (6.8.2+dfsg-7) ...
Setting up qml6-module-qtquick-controls:amd64 (6.8.2+dfsg-7) ...
Setting up qml6-module-qtquick-dialogs:amd64 (6.8.2+dfsg-7) ...
Setting up qml6-module-qt-labs-platform:amd64 (6.8.2+dfsg-7) ...
Setting up riseup-vpn (0.24.10+ds1-1+deb13u2) ...
Processing triggers for gnome-menus (3.36.0-3) ...
Processing triggers for libc-bin (2.41-12+deb13u4) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for mailcap (3.74) ...
Processing triggers for desktop-file-utils (0.28-1) ...
farwah@debian:~$ sudo apt update

```
farwah@debian:~$ riseup-vpn
No systray icon available.
qml: flavor: riseup-vpn
QSystemTrayIcon::setVisible: No Icon set
2026-11-10T14:15:24-05:00 INF Enabling logging hint="you can change the log leve
l with env LOG_LEVEL=INFO|DEBUG|TRACE" logFile=/home/farwah/.config/leap/systray
.log
2026-11-10T14:15:24-05:00 INF Not autostarting OpenVPN
2026-11-10T14:15:25-05:00 INF Checking for updates url=https://downloads.leap.se
/RiseupVPN/linux/lastver
2026-11-10T14:15:25-05:00 INF Sucessfully checked if there is an update updateAv
ailable=false
2026-11-10T14:15:26-05:00 INF Stopping firewall
2026-11-10T14:15:26-05:00 INF Using transport openvpn
2026-11-10T14:15:26-05:00 INF Setting transport transport=openvpn
2026-11-10T14:15:26-05:00 INF Enabled UDP
2026-11-10T14:15:26-05:00 INF Fetching gateways...
qml: show motd
qml: configured locale: en
qml: platform: linux
qml: status off
qrc:/components/BoldLabel.qml:15:5: Unable to assign [undefined] to QString
2026-11-10T14:15:50-05:00 INF Getting OpenVPN client certificate
qml: status off
qml: status off
2026-11-10T14:16:56-05:00 WRN Could not fetch geolocation error="Post \"https://
api.black.riseup.net:9001/json\": context deadline exceeded (Client.Timeout exce
eded while awaiting headers)"
2026-11-10T14:17:21-05:00 WRN Could not fetch geolocation error="Post \"https://
api.black.riseup.net:9001/json\": dial tcp 198.252.153.67:9001: i/o timeout (Cli
ent.Timeout exceeded while awaiting headers)"
qml: status starting
2026-11-10T14:18:51-05:00 WRN Could not fetch geolocation error="Post \"https://
api.black.riseup.net:9001/json\": context deadline exceeded (Client.Timeout exce
eded while awaiting headers)"
2026-11-10T14:18:51-05:00 INF Got best gateway [{vpn10-mtl.riseup.net 199.58.83.
11 Montreal Montreal CA [53 80 1194] [tcp udp] map[] openvpn} {vpn23-mia.riseup.
net 89.187.173.174 Miami Miami US [53 80 1194] [tcp udp] map[] openvpn} {vpn18-m
tl.riseup.net 199.58.83.9 Montreal Montreal CA [53 80 1194] [tcp udp] map[] open
vpn}]
2026-11-10T14:18:51-05:00 INF Starting firewall
2026-11-10T14:18:51-05:00 INF Starting OpenVPN
```

When you open the app




The meaning of the colors

- **Green:** The VPN tunnel is active and working.
- **Yellow:** The connection is unstable or currently connecting.
- **Red:** Disconnected or experiencing an issue.

The connection was unsuccessful since it was unstable and the school network is blocking it.

# 5. Team Member Contributions

### Kaci Luu

I worked on the PowerPoint slides, structuring all the content to be added in each slide. I also installed riseupVPN on my windows laptop to test out the features on it. The riseupVPN didn't successfully connect since the screen has been yellow the entire class time but at least I know now how RiseupVPN works. I also did research about VPN components and filled out some slides in the PowerPoint.

### Jia-Yu Joy Ho

I created the Github repo and worked on creating a temporary Wireguard setup using my SSD running a Debian OS. Although it wasn't functional (I believe it's due to handshake failure), this setup allowed me to familiarize with the commands and the Wireguard protocol. I will be able to recreate a working VPN tunnel using my home server after my experience with Wireguard today.

### Farwah Bakhsh

I worked on the PowerPoint presentation, specifically the slides on TailScale (basic introduction and how it works). I also mainly focused on researching about TailScale and RiseUp VPN, and how we could integrate them with WireGuard for our project. For this lab report document, I created the document and installed RiseUp VPN on Linux to showcase how the installation works. Also, on my Debian OS, the RiseUp VPN spent the whole class on 'Connecting', so it wasn't able to successfully connect at all.

# 6. Future Work Planned

*The list below is not in order.*

- Complete all the slides for the PowerPoint
- Complete the repository
- Configure the hardware / homelab for the VPN hosting
- Establish how to connect WireGuard, TailScale and RiseUp VPN together for the scope of our project.
- Try to host our own VPN in another server.
- Test out our VPN, making sure it is accessible from our devices.